Institución Universitaria
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Knowledge Distillation

INTEGRACIÓN DE ML EN EMBEBIDOS Y EDGE COMPUTING

Somos Innovación Tecnológica con Sentido Humano

Alcaldía de Medellín

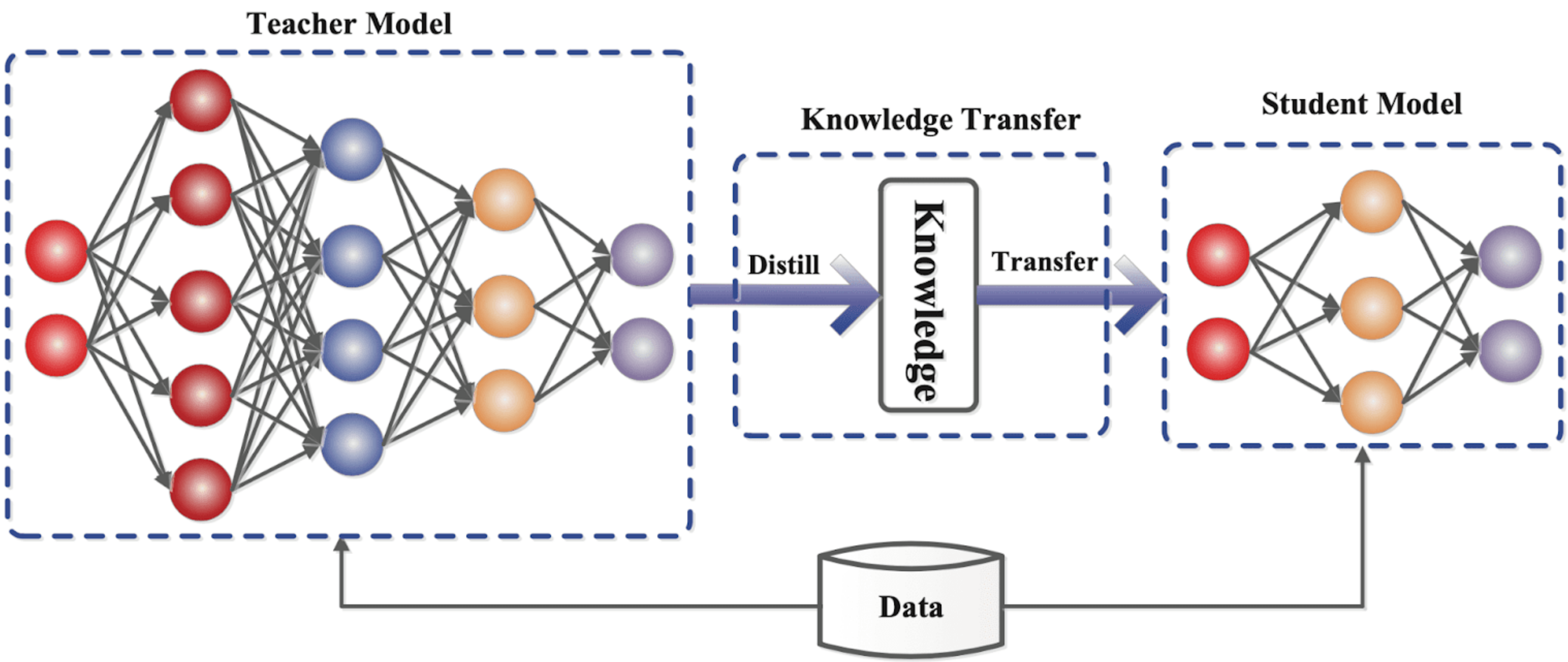
Outline

1. Introduction
2. KD Types
3. KD Algorithms
4. What to match?

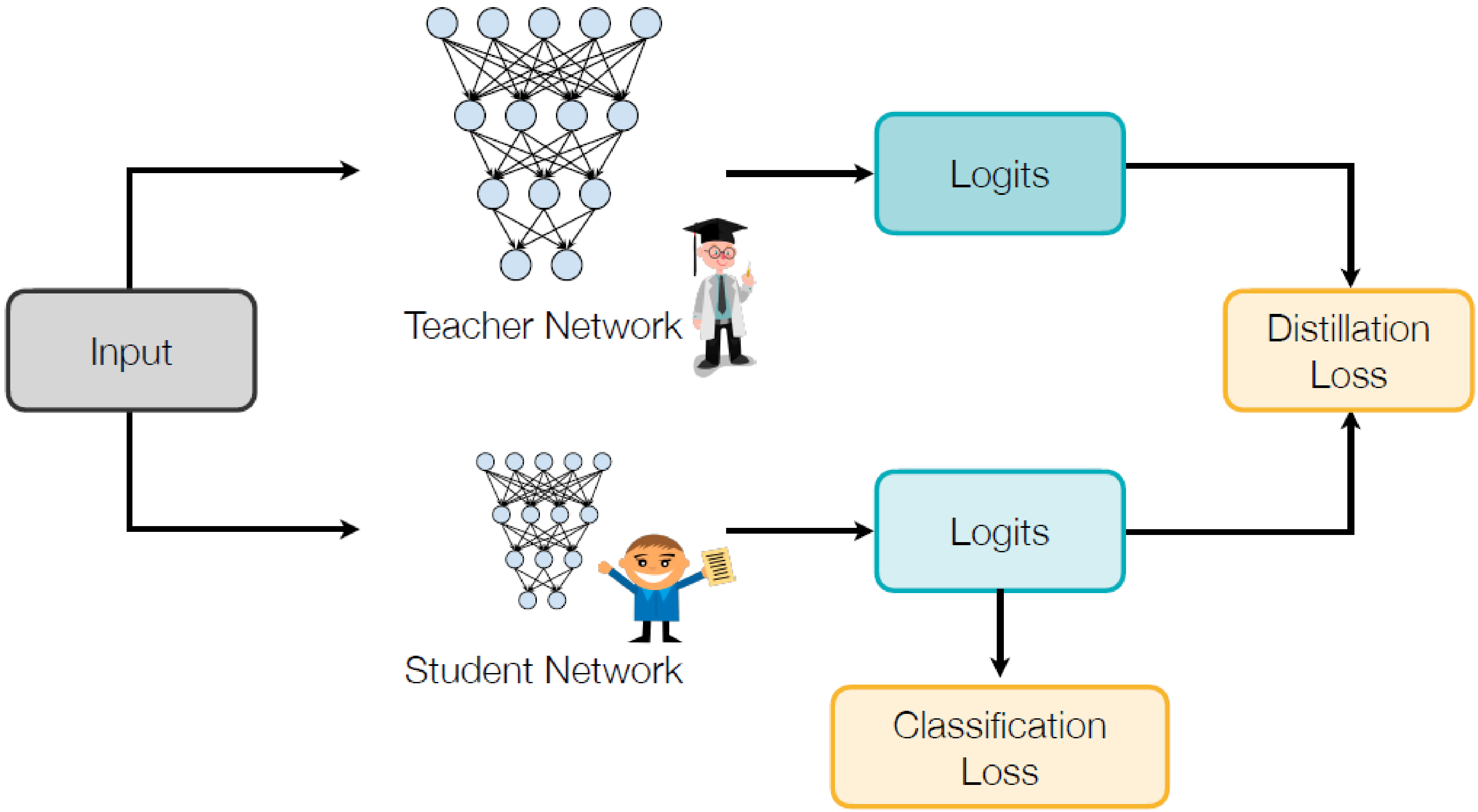
Concept

Knowledge distillation is used to compress a complex and large neural network into a smaller and simpler one, while still retaining the accuracy and performance of the resultant model.

Knowledge Distillation

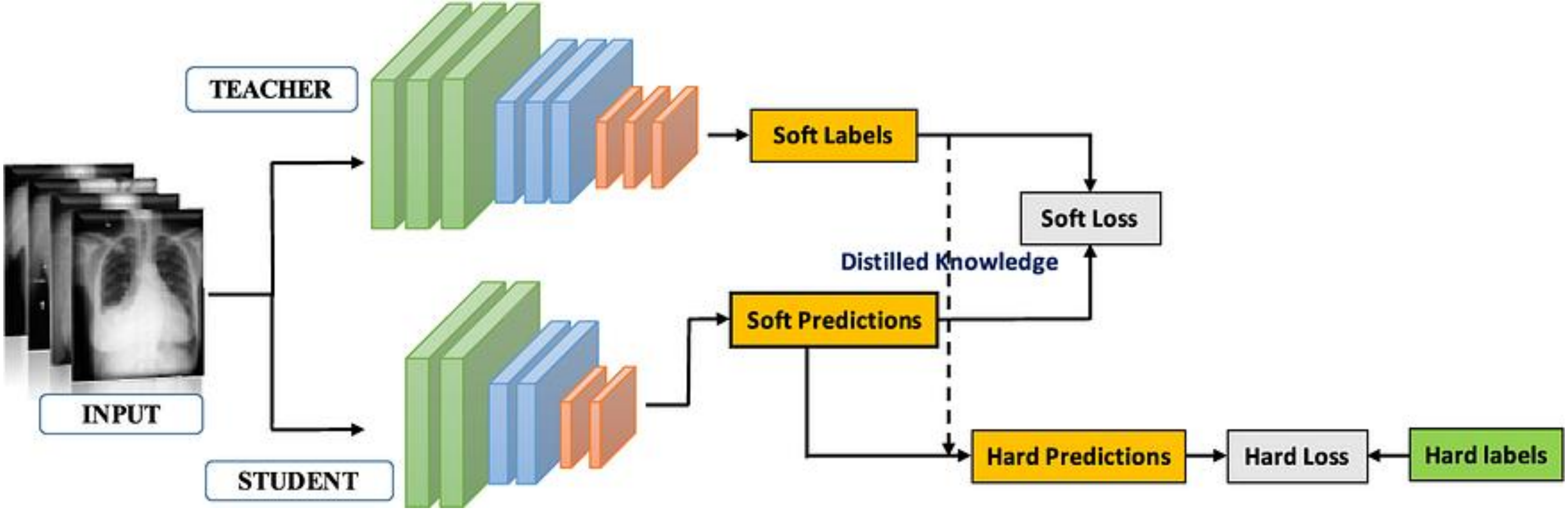


Knowledge Distillation



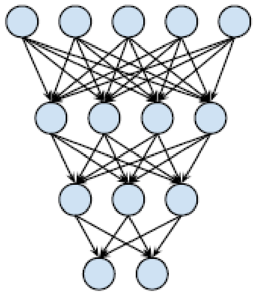
Distilling the Knowledge in a Neural Network [Hinton *et al.*, NeurIPS Workshops 2014]

Knowledge Distillation



Knowledge Distillation

Matching prediction probabilities between teacher and student

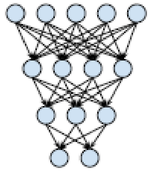


Teacher Network

	Logits	Probabilities
Cat	5	0.982
Dog	1	0.017

$$\frac{\exp(5)}{\exp(5)+\exp(1)}$$

$$\frac{\exp(1)}{\exp(5)+\exp(1)}$$



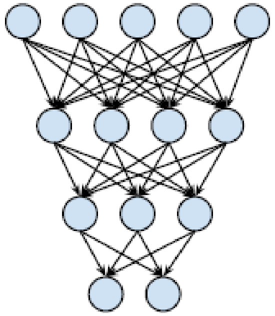
Student Network

	Logits	Probabilities
Cat	3	0.731
Dog	2	0.269

The student model is less confident

Knowledge Distillation

Concept of temperature



Teacher Network

$$\frac{\exp(5/1)}{\exp(5/1)+\exp(1/1)}$$

	Logits	Probabilities (T=1)	Probabilities (T=10)
Cat	5	0.982	0.599
Dog	1	0.017	0.401

$$\frac{\exp(5/10)}{\exp(5/10)+\exp(1/10)}$$

A larger temperature smooths the output probability distribution.

Soft targets

These soft labels are more informative than hard labels and capture the uncertainty and ambiguity in the predictions of the teacher network.

An example of hard and soft targets

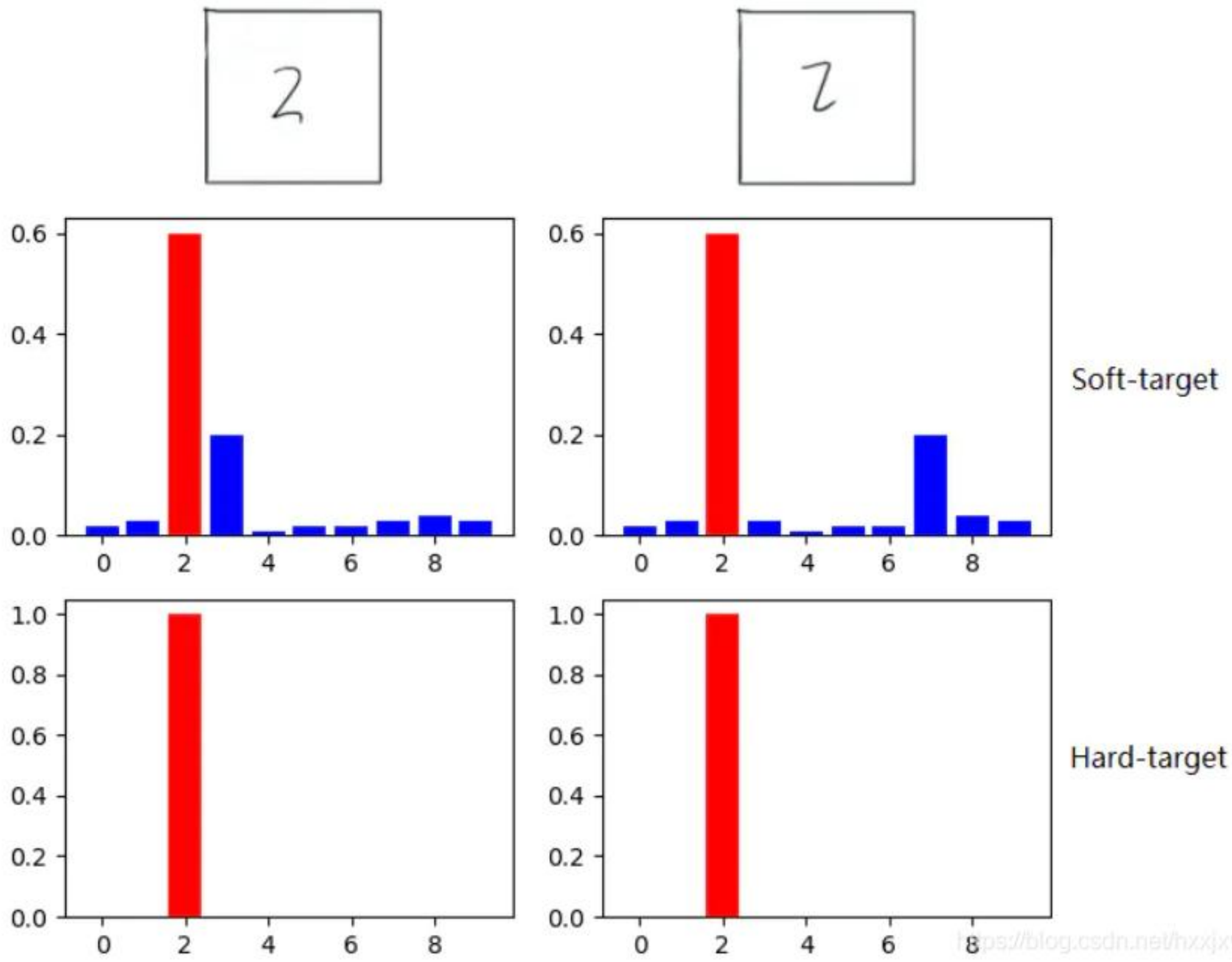
cow	dog	cat	car	
0	1	0	0	original hard targets

cow	dog	cat	car	
10^{-6}	.9	.1	10^{-9}	output of geometric ensemble

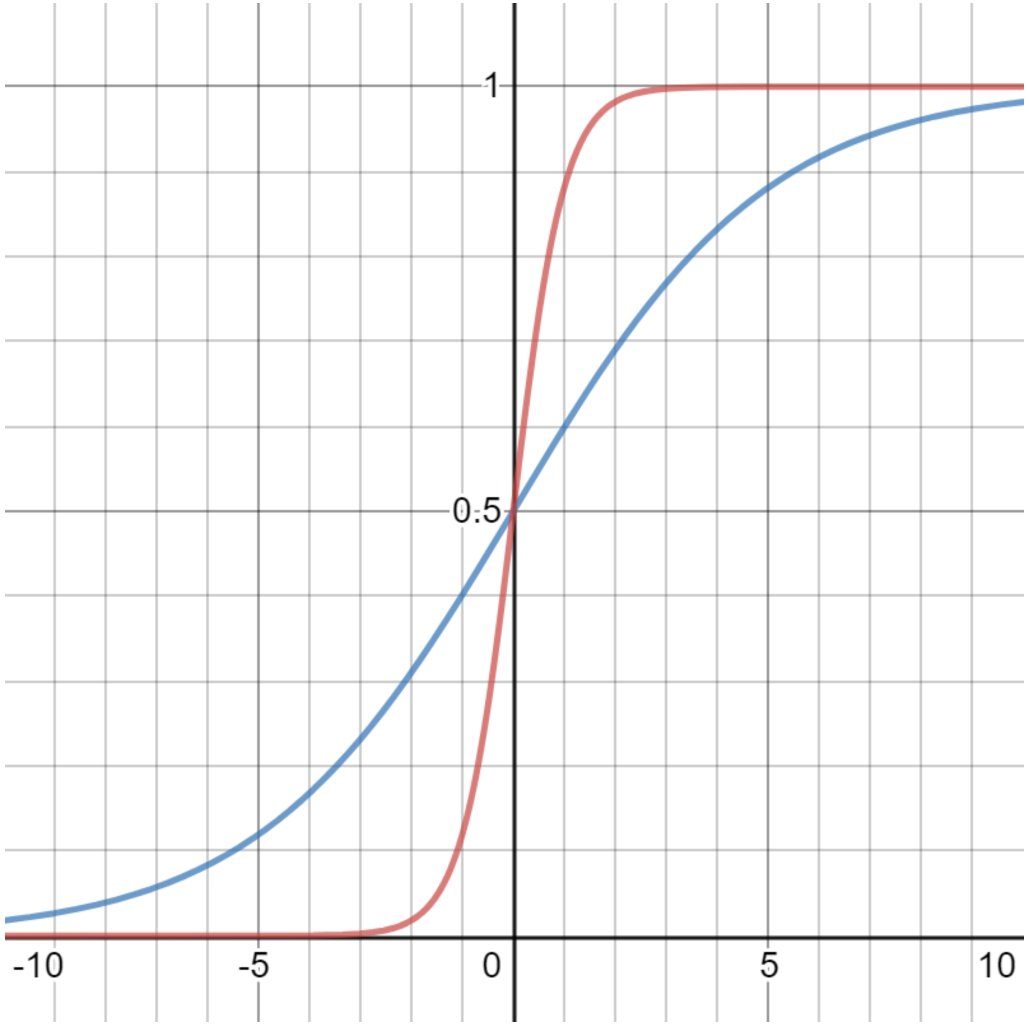
cow	dog	cat	car	
.05	.3	.2	.005	softened output of ensemble

Softened outputs reveal the dark knowledge in the ensemble.

Soft targets



Loss for Soft Targets



In higher temperature(T) settings, the class scores change more smoothly, and larger amount of knowledge will be transferred.

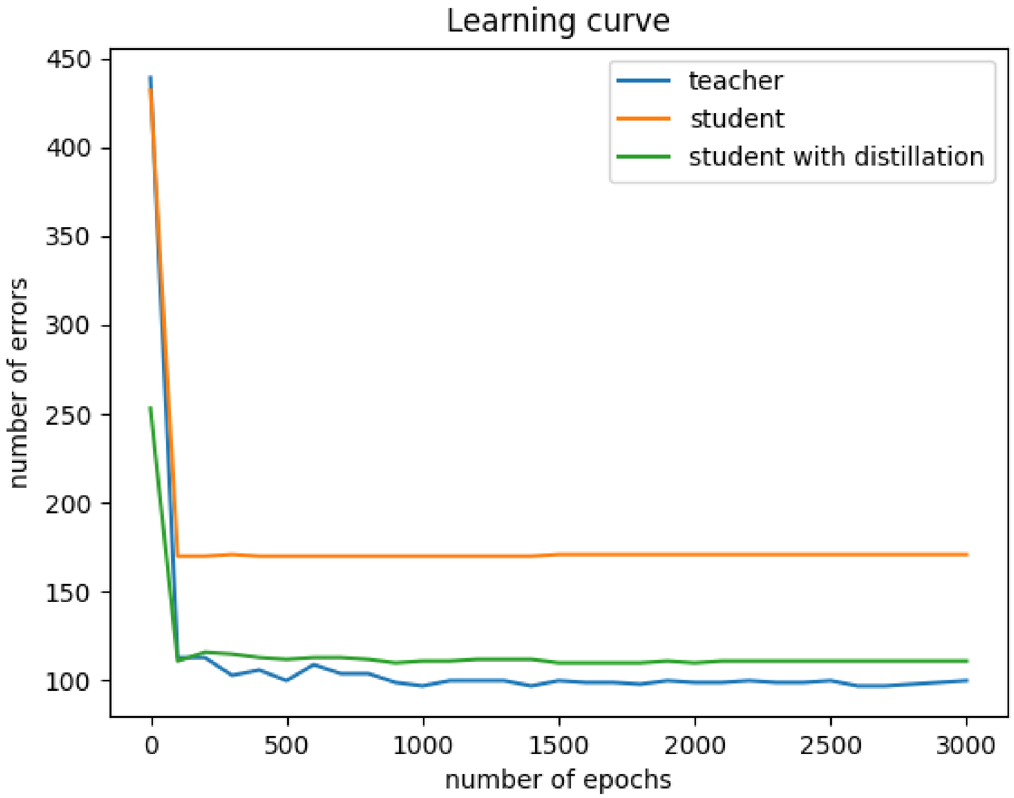
$$q_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

q_i : class score

z_i : logit

T : Temperature

Example



The goal of knowledge distillation is to **align the class probability distributions from teacher and student networks**.

- Teacher:
  784 → ReLU → 1200 → ReLU → 1200 → 10
  (dropout 20% of input, and 80% of the two hidden layer's output)
- Student:
  784 → ReLU → 800 → ReLU → 800 → 10

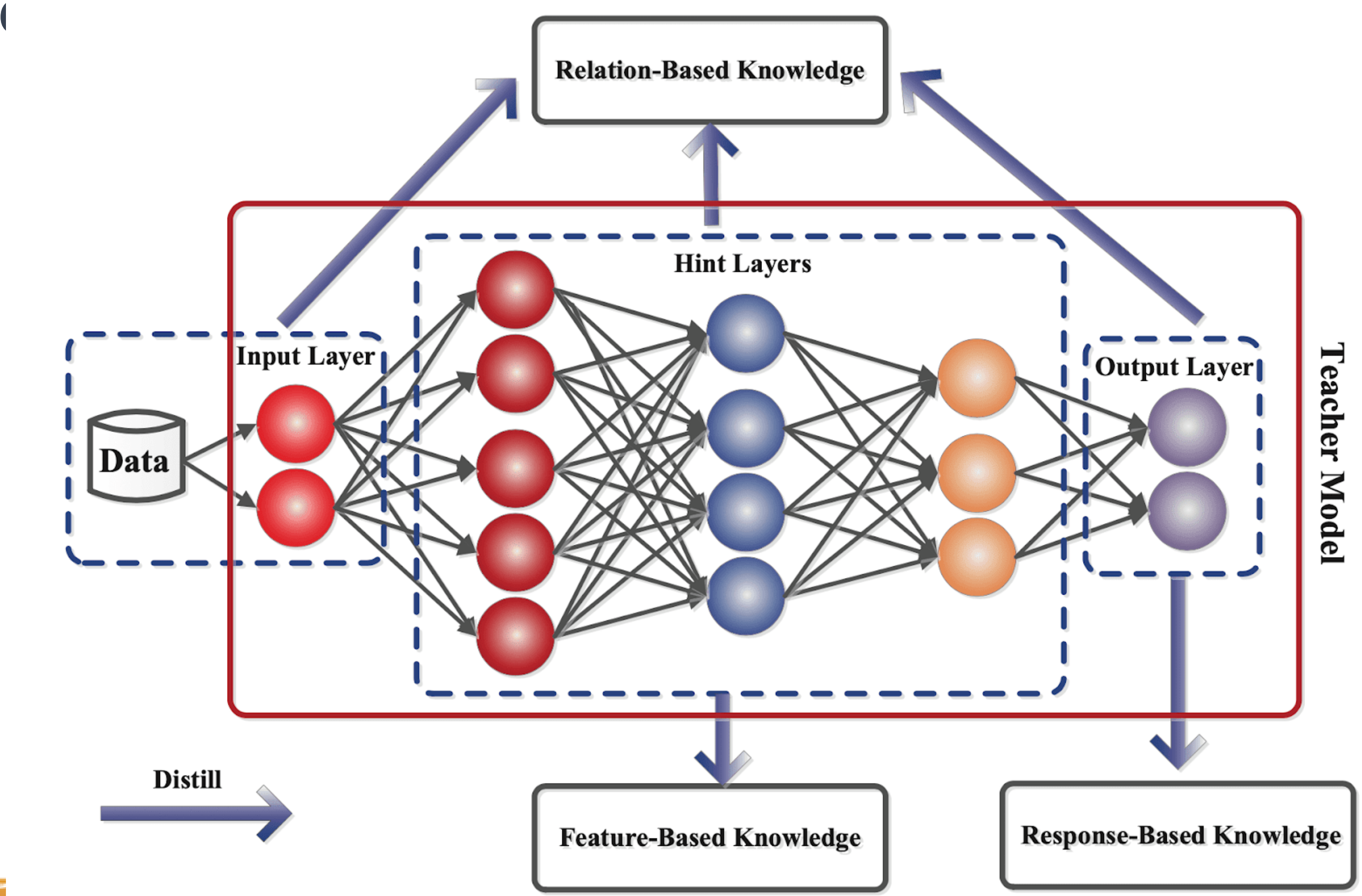
Outline

1. Introduction
2. DK Types
3. DK Algorithms
4. What to match?

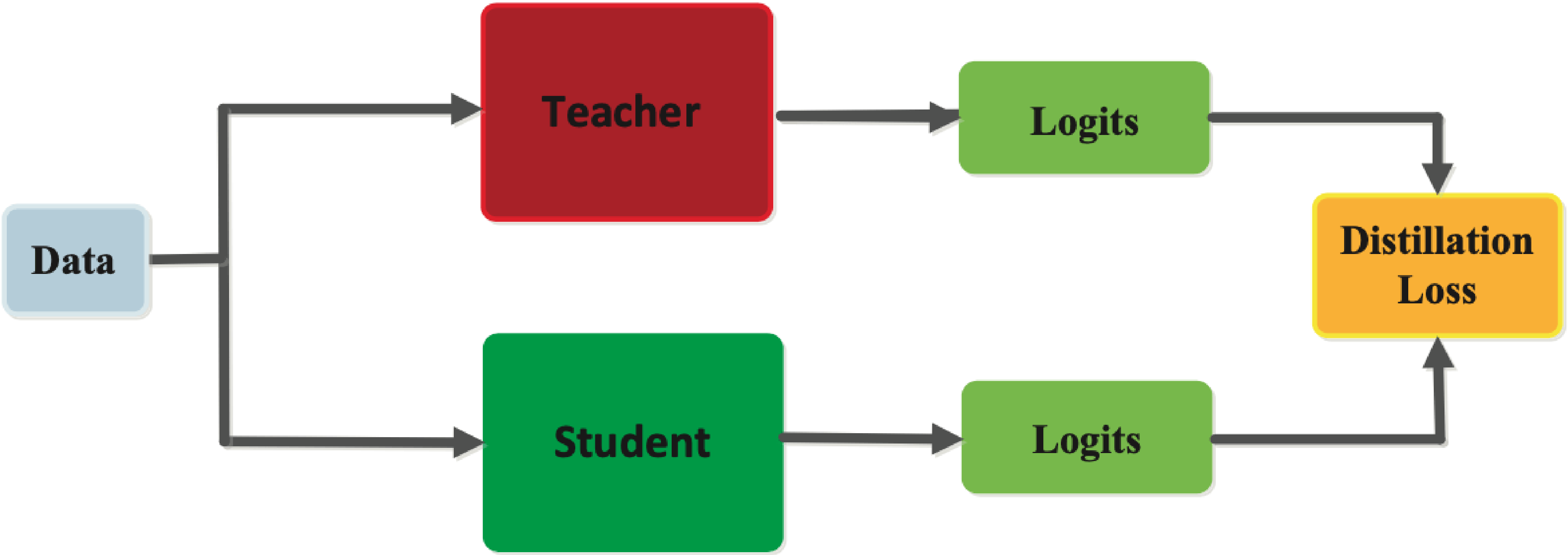
KD Types

Generally, there are three types of knowledge distillation, each with a unique approach to transferring knowledge from the teacher model to the student model. These includ

- Response-based distillation
- Feature-based distillation
- Relation-based distillation



Response-based distillation



This technique only transfers knowledge related to the teacher model's predicted outputs and does not capture the internal representations learned by the teacher model.

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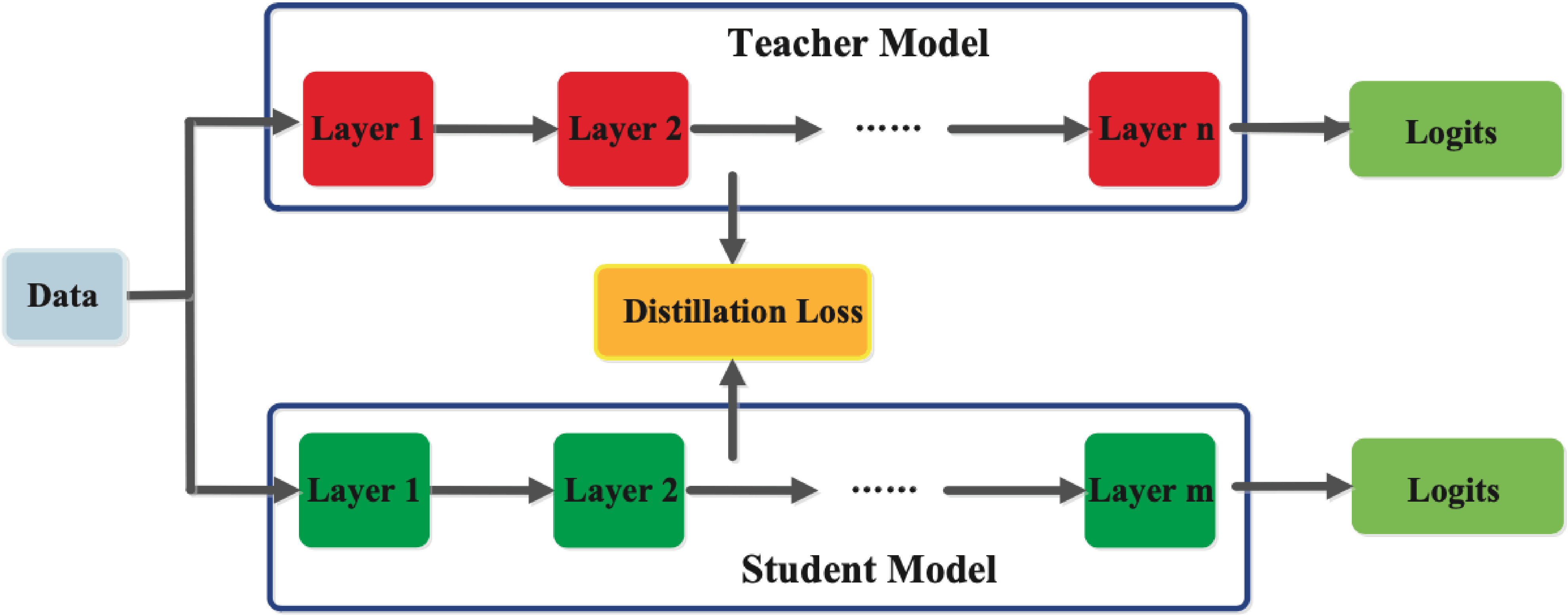
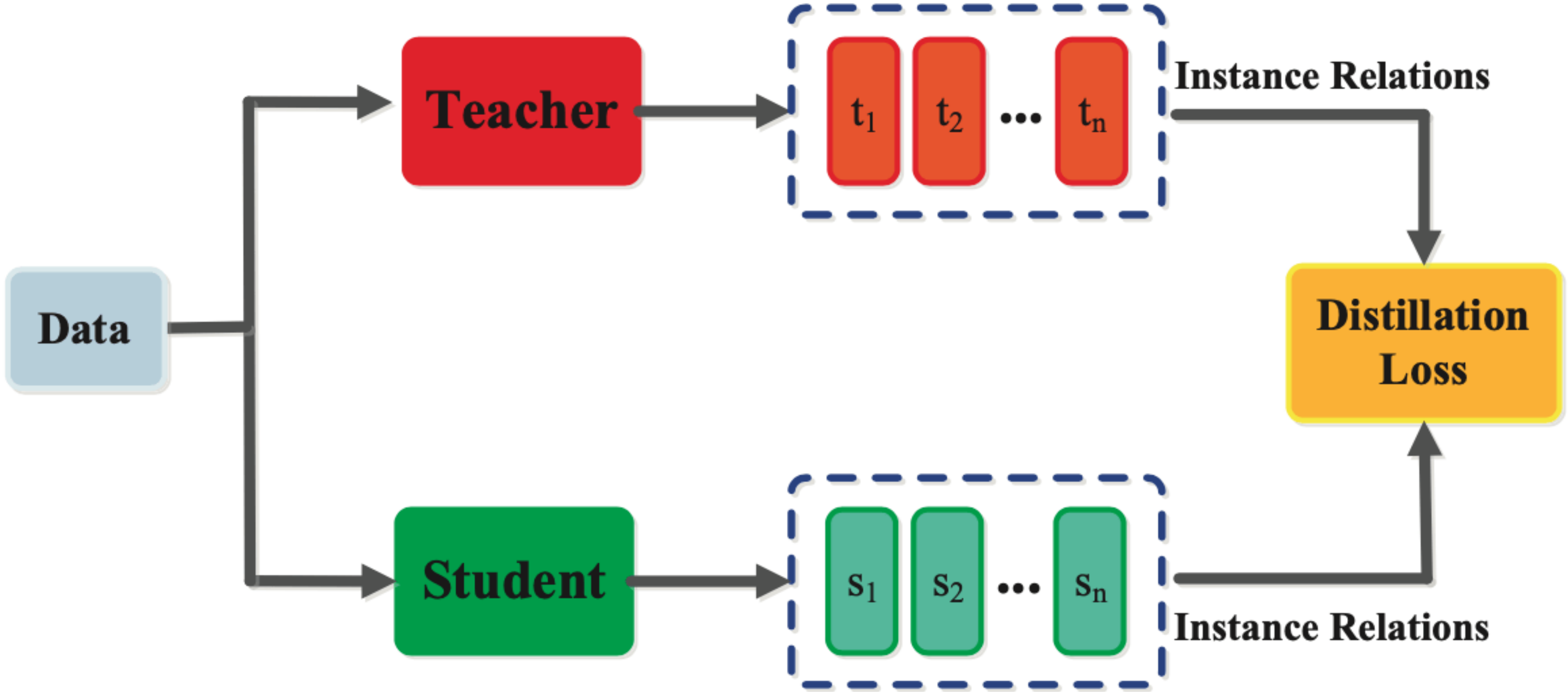
Feature-based distillation



This technique can be more computationally expensive than other types of knowledge distillation, as it requires extracting the internal representations from the teacher model at each iteration

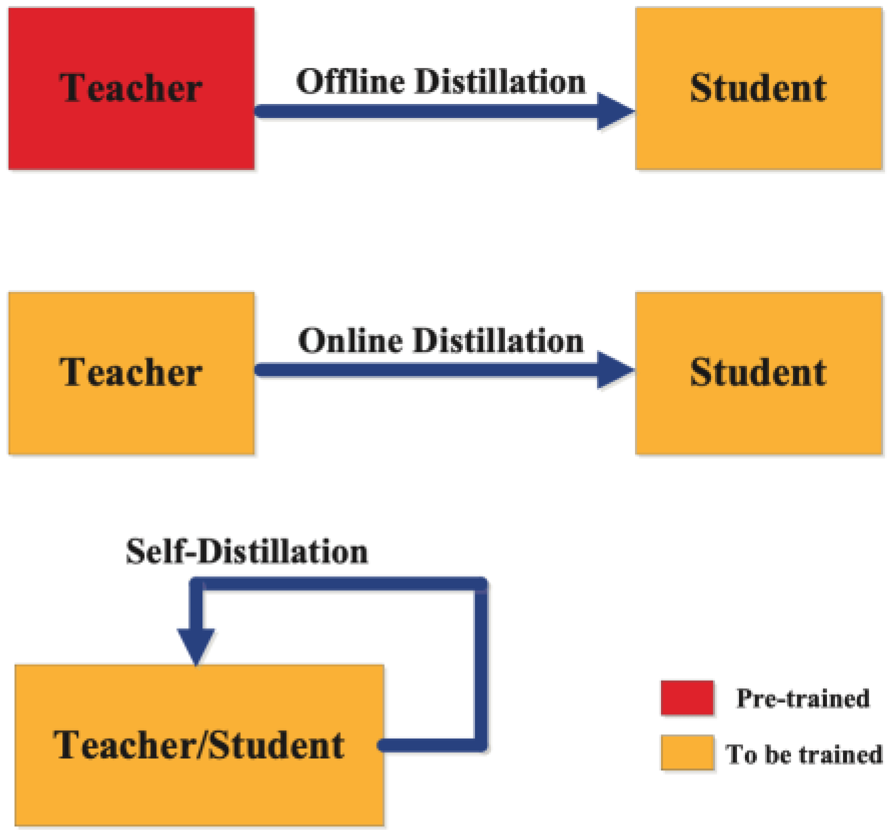
Relation-based distillation



A student model is trained to learn a relationship between the input examples and the output labels. Relation-based distillation focuses on transferring the underlying relationships between the inputs and outputs

Training

The categorization of the distillation training methods depends on whether the teacher model is modified at the same time as the student model or not.



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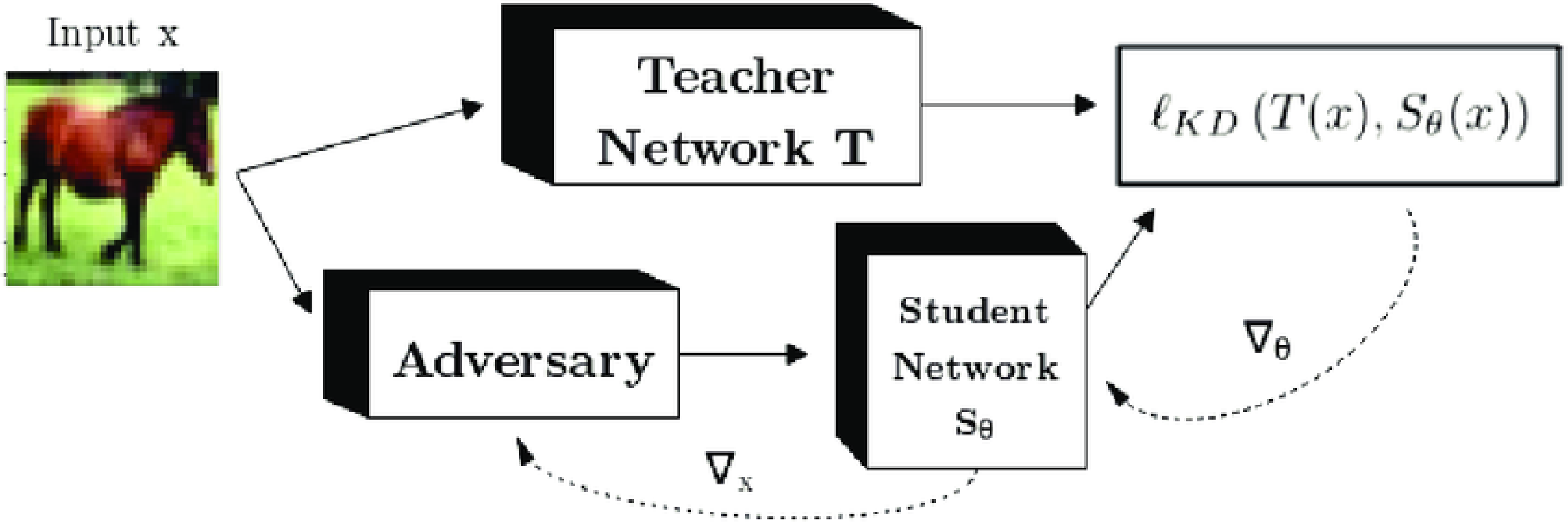
KD Algorithms

Specifically, we will discuss three algorithms:

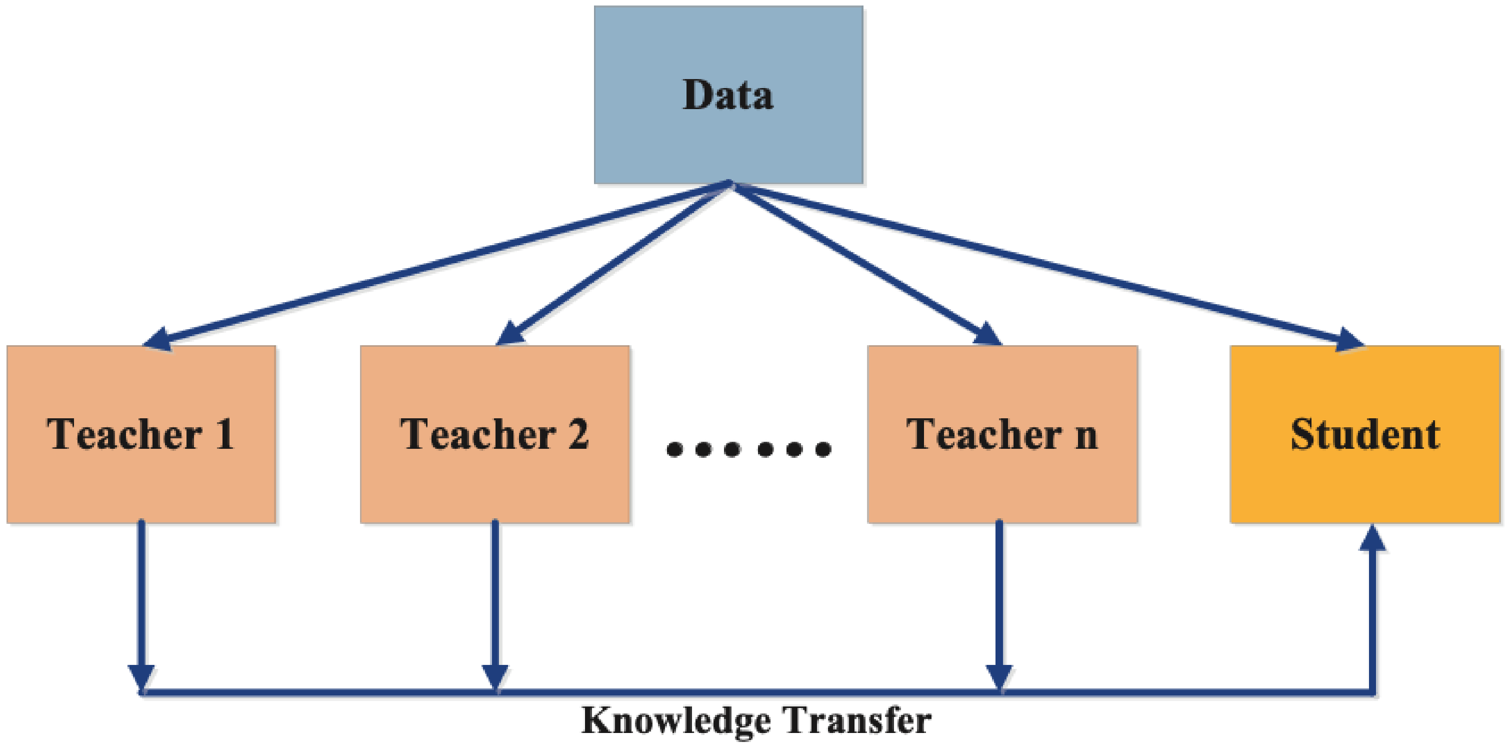
1. Adversarial distillation
2. Multi-teacher distillation
3. Cross-modal distillation

Adversarial distillation



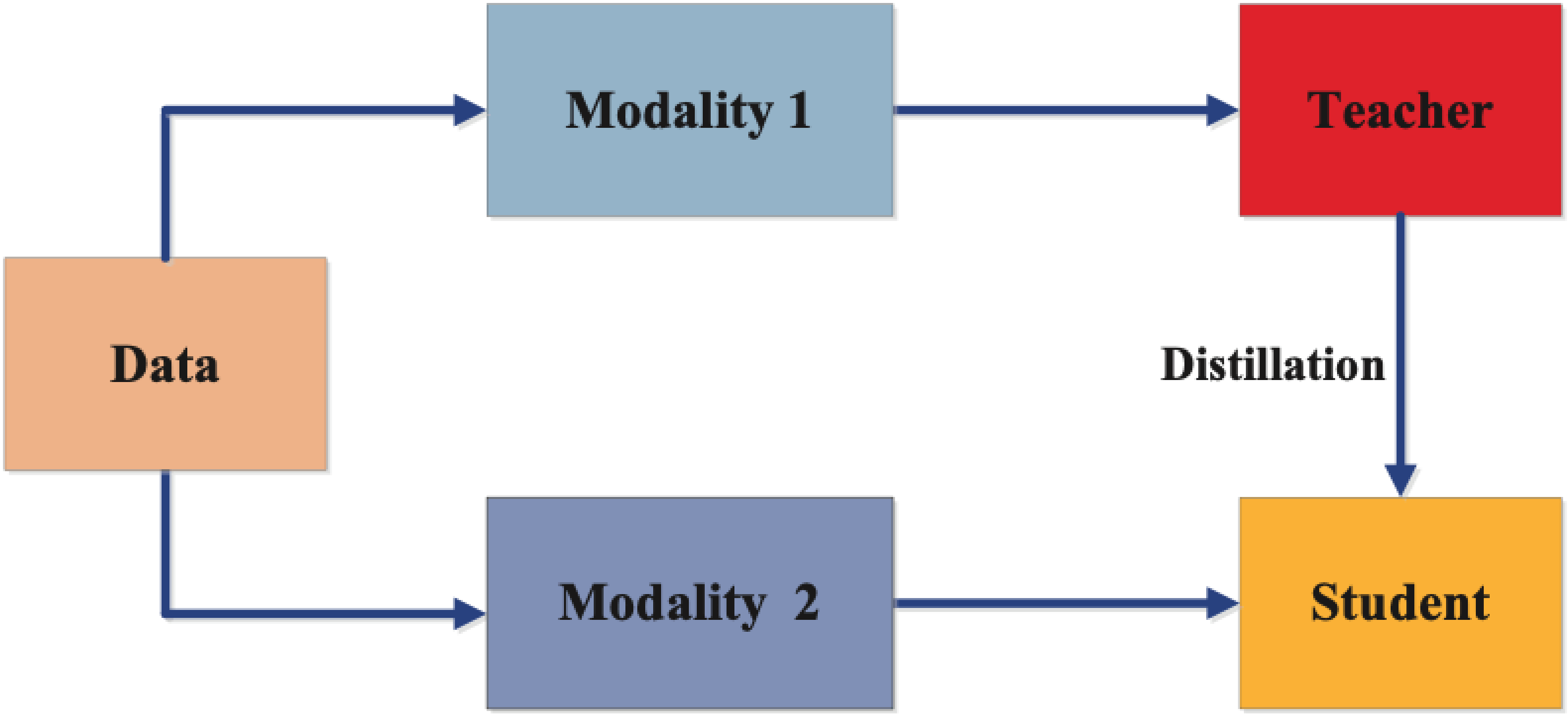
The adversarial network is trained to generate samples that the teacher model finds difficult to classify correctly, and the student model is trained to classify these samples correctly.

Multi-Teacher Distillation



Multi-teacher distillation has several advantages over traditional knowledge distillation techniques. This technique reduces the bias that can be introduced by a single teacher model, as multiple teachers provide a more diverse set of perspectives.

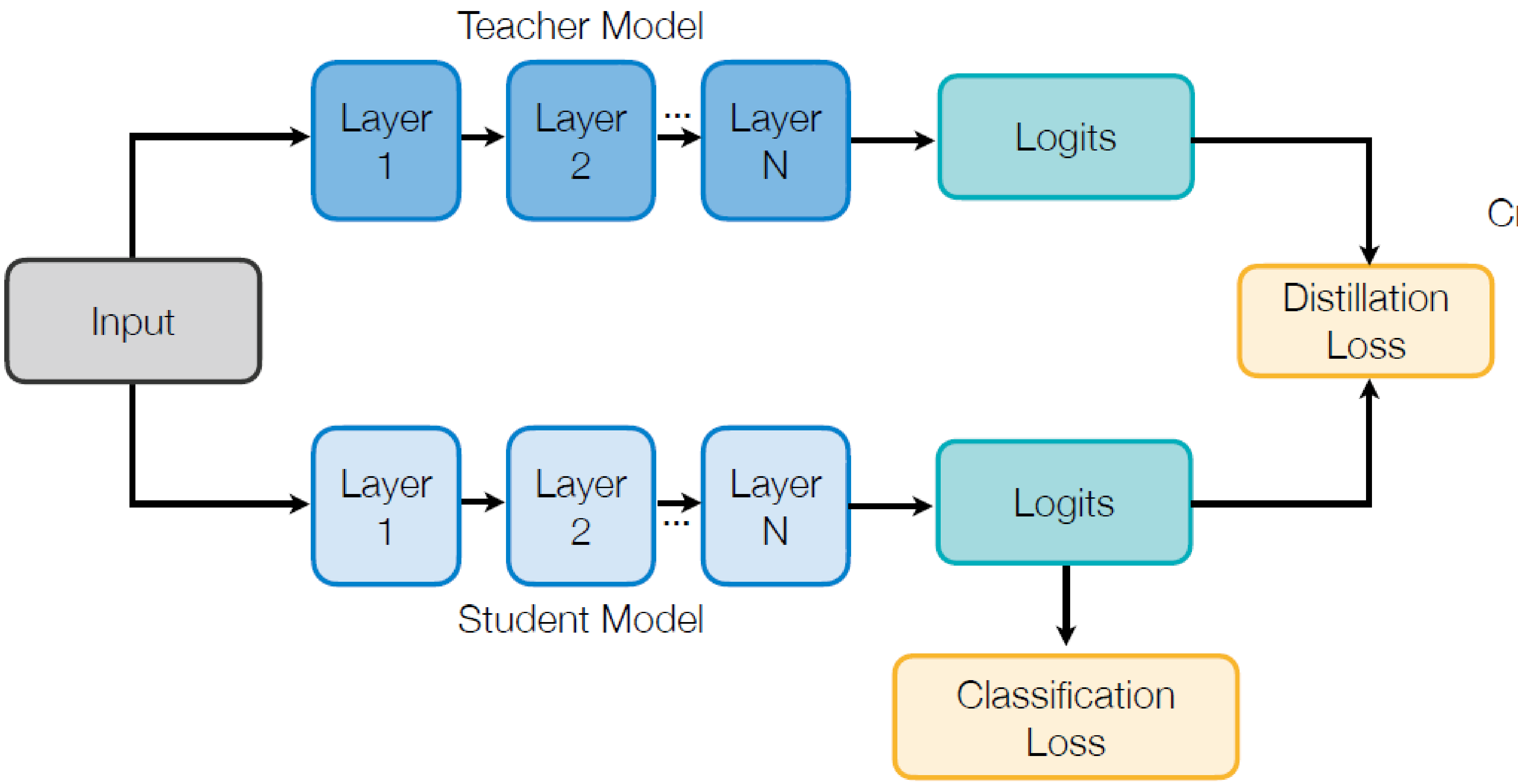
Cross-Modal Distillation



Cross-modal distillation allows the transfer of knowledge from one modality to another, making it useful when data is available in one modality but not in another. Furthermore, cross-modal distillation improves the generalization capability of the student model, as it learns from a more comprehensive set of features.

Outline

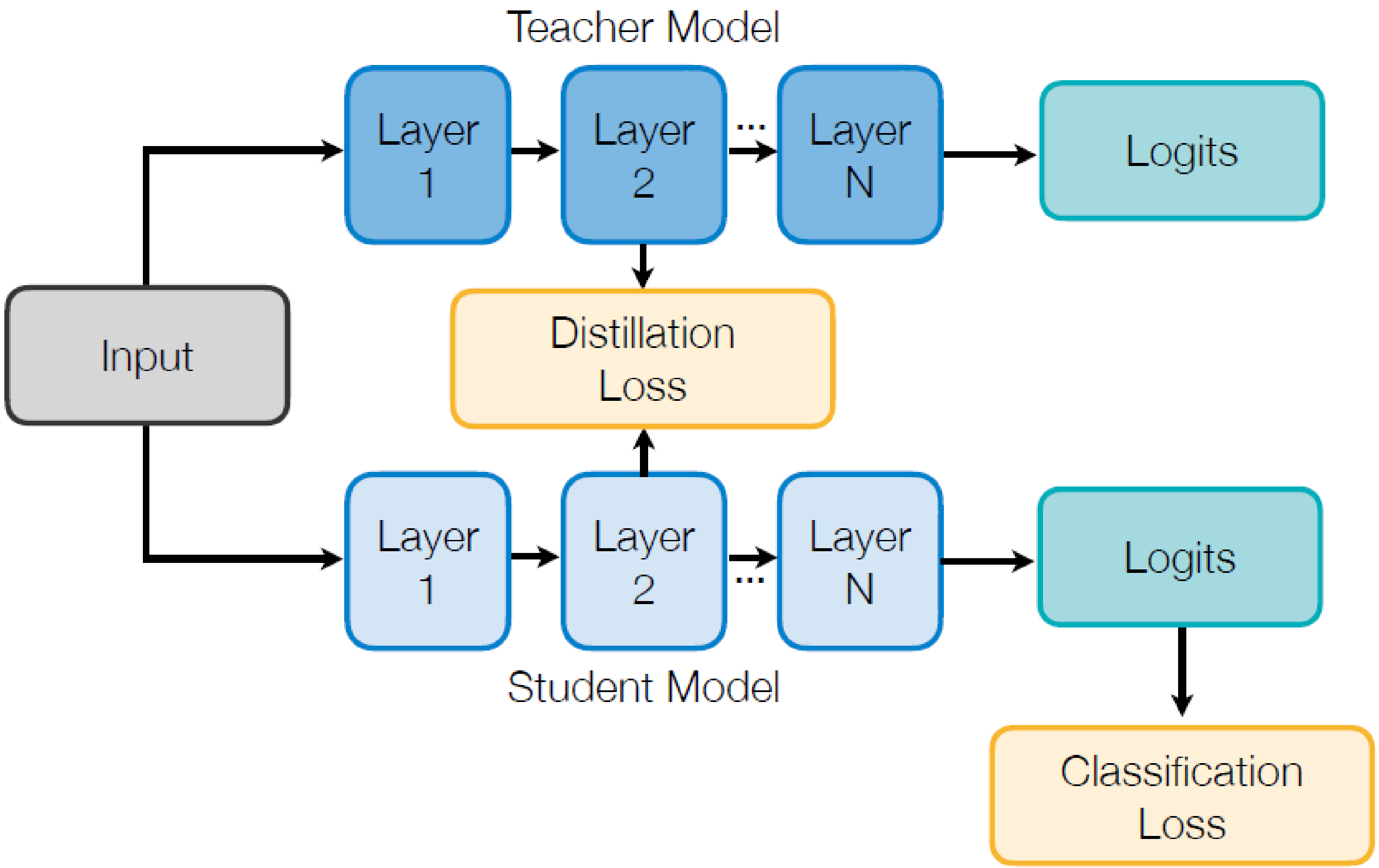
Matching output logits



Cross entropy loss:
$\mathbb{E}(-p_t \log p_s)$;

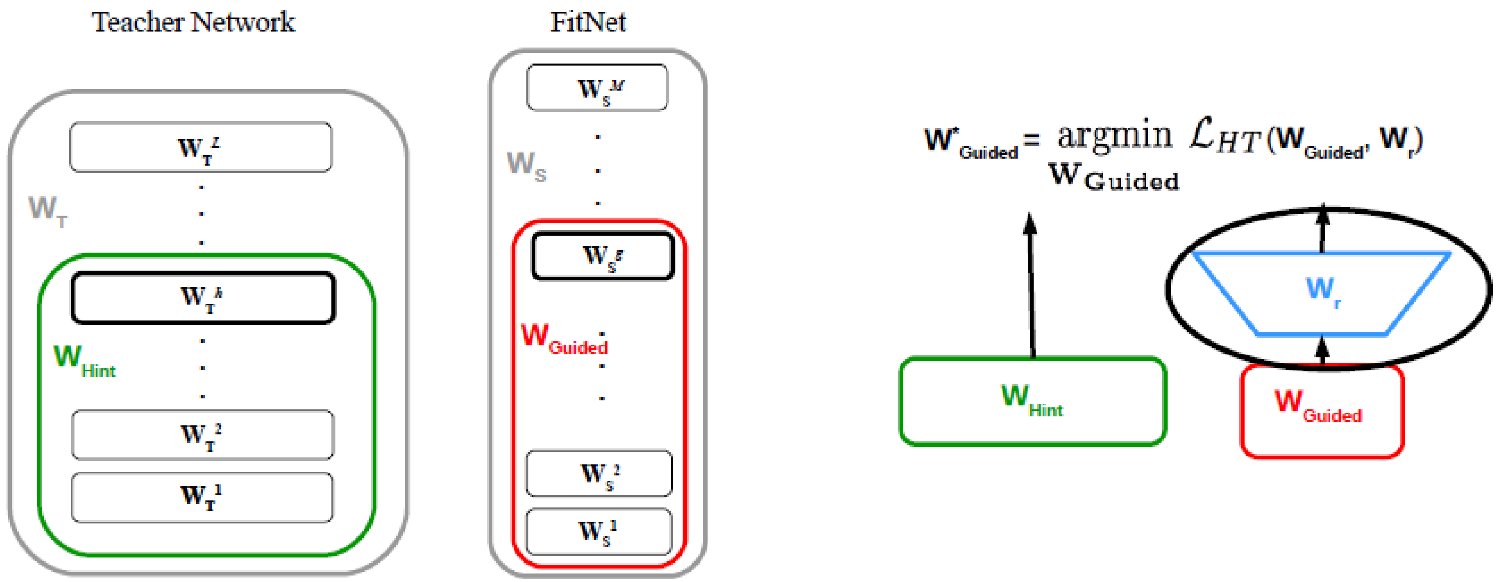
L2 loss:
$E(\|p_t - p_s\|_2^2)$

Intermediate weights



Knowledge Distillation: A Survey [Gou *et al.*, IJCV 2020]

Intermediate weights



(a) Teacher and Student Networks

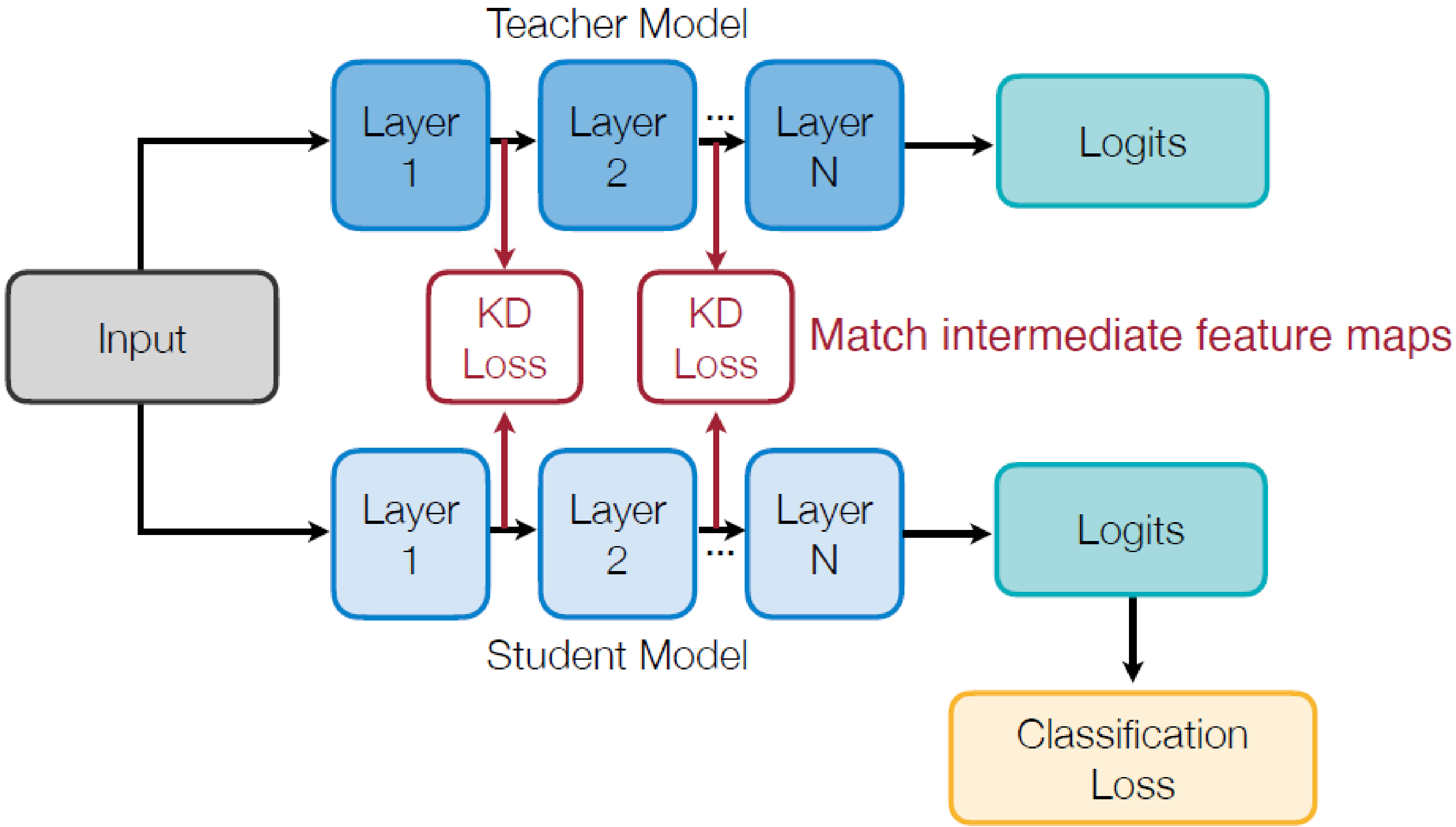
(b) Hints Training

An FC layer used to align the shapes of teacher and student weights

Other than the cross-entropy distillation loss, also add a L2 loss between teacher weights and student weights (linear transformation is applied to match the dimensionalities).

FitNets: Hints for Thin Deep Nets [Romero *et al.*, ICLR 2015]

Intermediate Features



Minimizing maximum mean discrepancy between feature maps

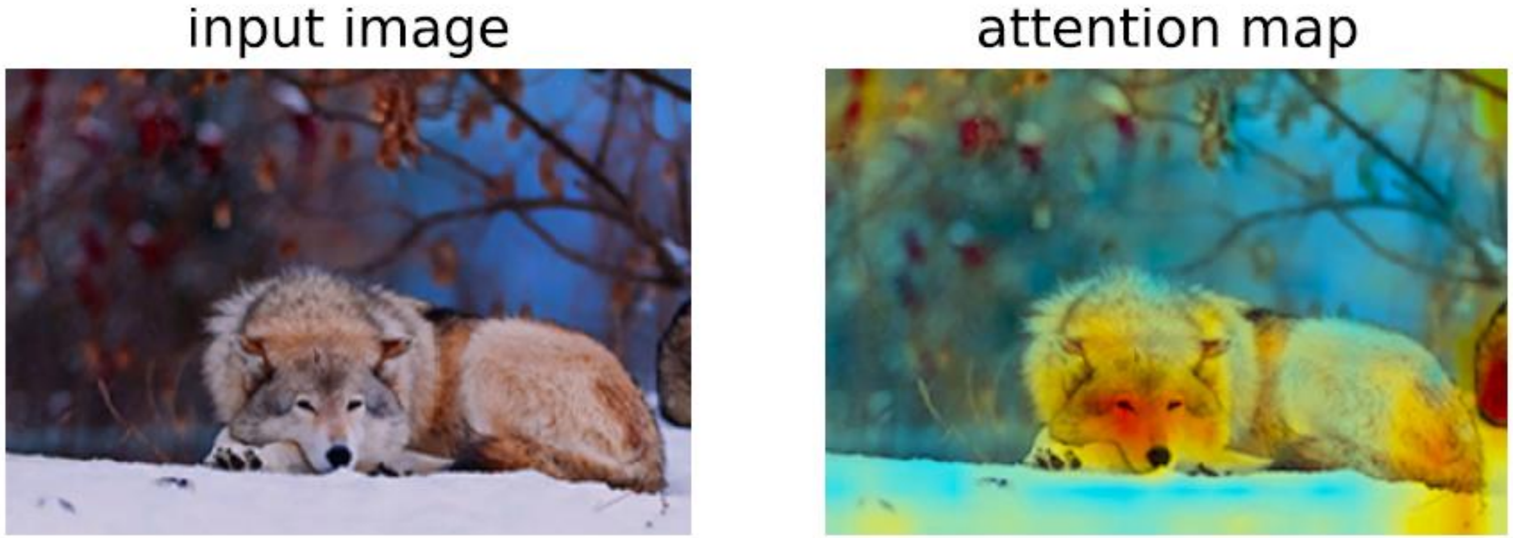
Teacher and student networks should have similar **feature** distributions, not just output probability distributions.

Like What You Like: Knowledge Distill via Neuron Selectivity Transfer [Huang and Wang, arXiv 2017]

Gradients – Attention maps

Gradients of feature maps are used to characterize "attention" of DNNs

- The attention of a CNN feature map x is defined as $\dfrac{\partial L}{\partial x}$, where L is the learning objective.
- Intuition: If $\dfrac{\partial L}{\partial x_{i,j}}$ is large, a small perturbation at i, j will significantly impact the final output. As a result, the network is putting more attention on position i, j.

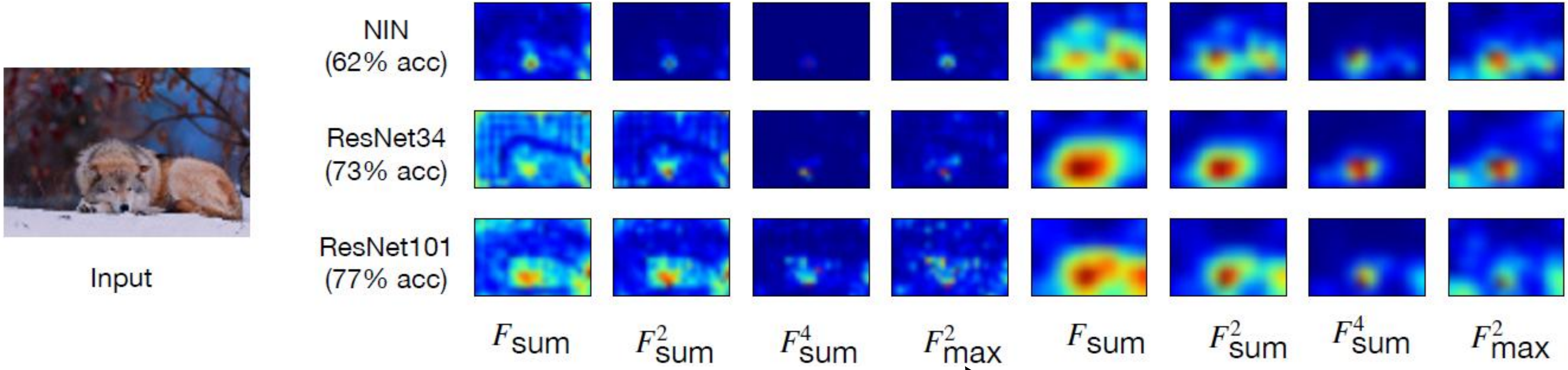


Paying More Attention to Attention: Improving the Performance of Convolutional Neural Networks via Attention Transfer [Zagoruyko and Komodakis, ICLR 2017]

Gradients – Attention maps

Performant models have similar attention maps

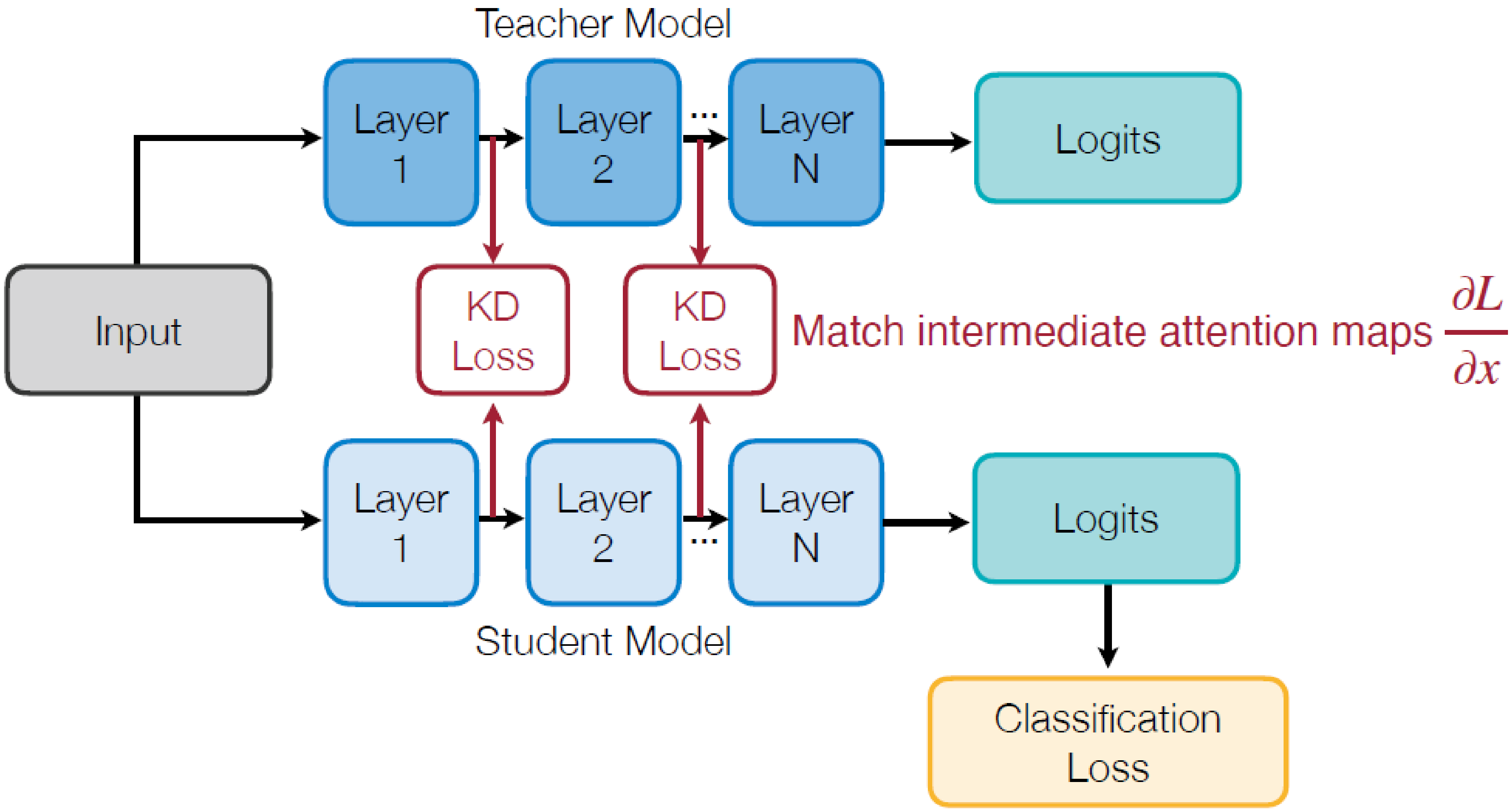
Attention maps of performant ImageNet models (ResNets) are indeed similar to each other, but the less performant model (NIN) has quite different attention maps.



Different reduction methods across the channel dimensions

Paying More Attention to Attention: Improving the Performance of Convolutional Neural Networks via Attention Transfer [Zagoruyko and Komodakis, ICLR 2017]

Intermediate attention maps



Paying More Attention to Attention: Improving the Performance of Convolutional Neural Networks via Attention Transfer [Zagoruyko and Komodakis, ICLR 2017]

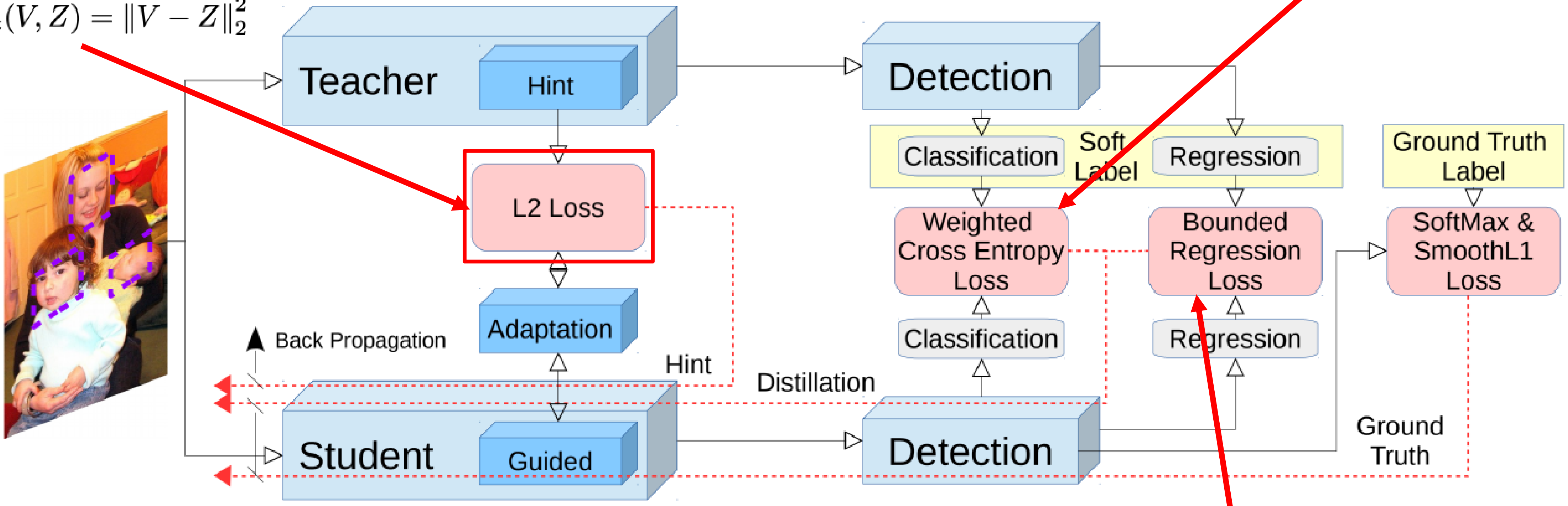
KD for Object Detection

Feature Imitation

Use different weights for foreground and background classes to handle the class imbalance problem.

$$L_{soft}(P_s, P_t) = -\sum w_c P_t \log P_s$$

$$L_{Hint}(V, Z) = \|V - Z\|_2^2$$

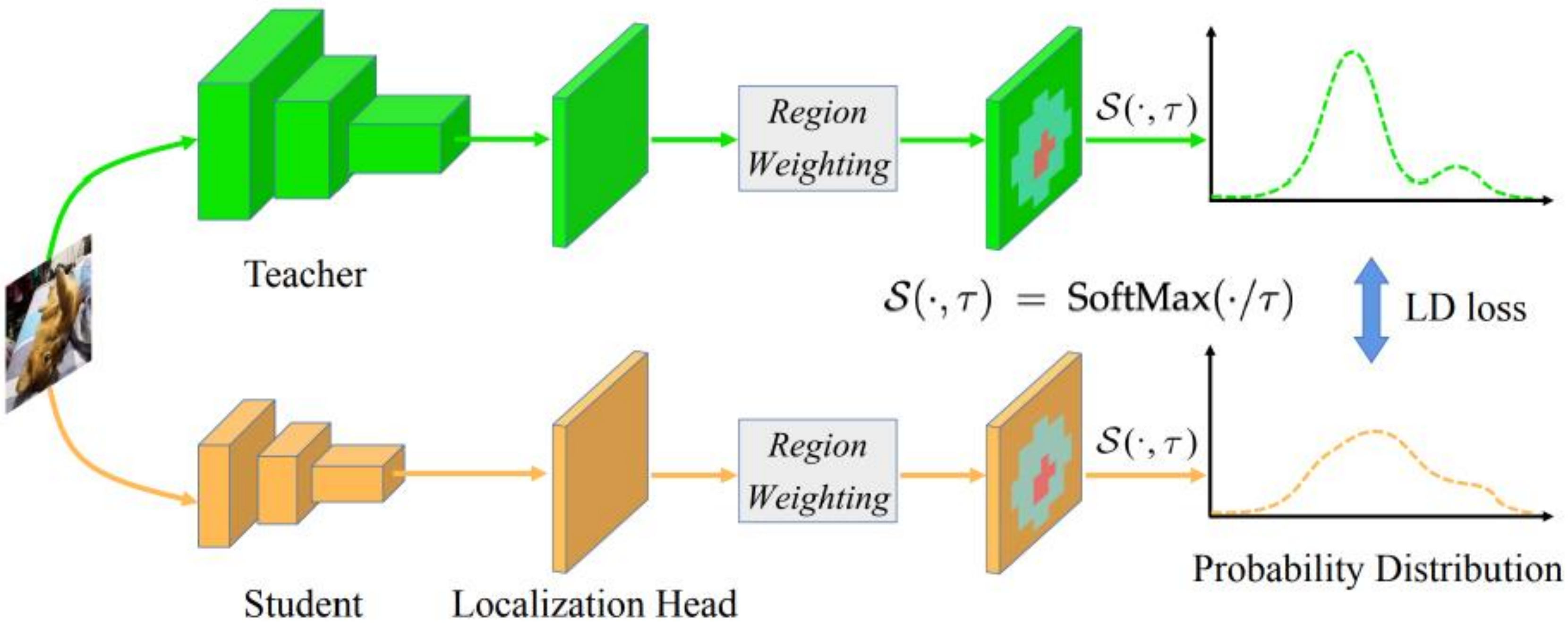


- Add a 1x1 convolution layer to match the shape.

$$L_b(R_s, R_t, y) = \begin{cases} \|R_s - y\|_2^2, & \text{if } \|R_s - y\|_2^2 + m > \|R_t - y\|_2^2 \\ 0, & \text{otherwise} \end{cases}$$

Learning Efficient Object Detection Models with Knowledge Distillation [Chen *et al.*, NeurIPS 2017]

KD for Object Detection



Calculate the distillation loss between two probability distributions predicted by the teacher and the student.

Localization Distillation for Dense Object Detection [Zheng *et al.*, CVPR 2022]

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